

Multiple Choice (10 marks)

1. Which is a valid expression in Python 3.5?
 - (a) `sort('ab')`
 - (b) `sorted('ab')`
 - (c) `"ab".sort()`
 - (d) `1/0`
2. Which of the following best explains what happens when a new device is connected to the Internet?
 - (a) A device driver is assigned to the device.
 - (b) An Internet Protocol (IP) address is assigned to the device.
 - (c) A packet number is assigned to the device.
 - (d) A Web site is assigned to the device.
3. Let `l = [1,2,3,4]`. What is `sum(l)` in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
4. In Python 3, what is `['a', 'Chemistry', 0, 1][-3]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
5. Which of the following spreadsheet functions would be most useful for detecting improbably high or low values that have become part of a data set as a result of data entry errors?
 - (a) A function that averages numeric values in a column or row
 - (b) A function that counts the values in a column or row
 - (c) A function that rounds a numeric value
 - (d) A function that sorts values in a column or row

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, using the syntax `b[::-2]` with `b = [11,13,15,17,19,21]` returns the list `[19,21]`.
7. True or False: $O(1)$ is the largest asymptotic complexity among the given functions.
8. True or False: In Python 3, the `seed([x])` function sets the starting value for generating random numbers.
9. True or False: In Python 3, the operation `x << 3`, where `x` is 1, results in the value 16.
10. True or False: In Python 3, the `replace()` function removes all leading and trailing whitespace from a string.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In Python 3, explain how the `'isupper()'` function works and analyze its importance in string manipulation. Provide examples of its use in validating user input.
12. Explain how the lambda function `r = lambda q: q * 2` works in Python 3, and calculate the output of `r(3)`. Discuss the implications of using lambda functions.

End of Paper